

A Data-Driven Look into Class Division: From Degrees to Demographics

Income inequality shapes the social and economic fabric of cities, raising the question: are struggling cities closing the gap, or being left further behind?

In this report, we analyze Canadian Census data to determine whether lower-income cities are falling further behind or catching up in terms of average income. We also investigate potential reasons behind these trends by examining whether selected socio-economic features (immigration, population, lone-parent household and degree) influence income levels. Our analysis includes statistical correlations, clustering, best-fit line modeling, and the use of random forest feature importance.

A significant part of this project involved cleaning and merging census data for 12 (total of 24 – 2015 to 2020) cities in British Columbia, then extracting the variables most relevant to our questions. This refined dataset allowed us to focus on interesting variables such as education levels, immigration rates, lone-parent household rates, and population changes when exploring patterns of income growth.

Understanding these relationships can provide insight into the dynamics of income inequality and help identify which factors may contribute to economic disparity.

Cleaning the data

For this project, Canadian census data at the census subdivision level was obtained for 12 selected cities. Data for the year 2020 (reported in 2021) and the same cities for the year 2015 (reported in 2016) were extracted to allow for a comparative analysis over time. Unfortunately, census data for 2025 is not yet available and is expected to be released in 2026, limiting the study to a two-point time frame.

The 24 datasets (CSV files) were in poor condition, with no clear structure or consistent formatting. Many contained duplicate column names and, in some cases, duplicate row labels. An additional complication was that the 2016 files did not follow the exact same format as the 2021 files, requiring a slightly modified version of the 2021 cleaning script to process them correctly. The scripts were written based on 1 data set from each year and then tested on the remaining 22.

Year-specific cleaning scripts were developed to handle differences between the 2016 and 2021 census formats, such as additional header rows, varying column names, and altered flag column patterns. Irrelevant thematic categories (e.g., language, religion, mobility) were removed using keyword matching on the Topic column, while non-data footer sections were trimmed by detecting the word “Symbols:” row. Duplicate and non-numeric columns were dropped, unnecessary “_Flag” columns were removed with regular expressions, and empty columns were eliminated entirely. This made looking through the dataset much easier. Then at last, I merged the two years into one dataset.

Techniques

To ensure comparability across cities of different sizes, the immigrant count, and lone-parent household count were normalized by each city's population, producing rates rather than raw counts. This prevented population size from disproportionately influencing the analysis.

To examine whether average after-tax income had genuinely increased between 2015 and 2020, a Mann–Whitney U test was performed. This non-parametric test was chosen because it does not assume a normal distribution of income and is well-suited to small sample sizes.

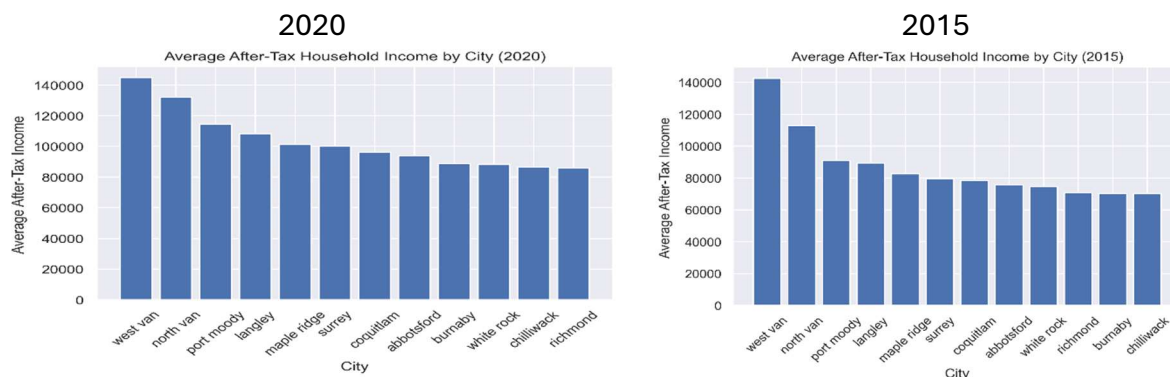
Relationships between education levels and income were explored using both linear and cubic regression models. A simple linear regression was applied to assess whether there was a statistically significant relationship between average income and bachelor's degree rate, with the p-value used as evidence of significance. A cubic polynomial fit was also generated, as visual inspection suggested it more closely followed the trend in the data.

To identify patterns in the relationship between income growth and growth in the number of degree holders, K-Means clustering was applied with $k=3$. This allowed cities to be grouped into clusters with similar growth rates, revealing patterns of socioeconomic development.

Finally, a Random Forest Regressor was used to calculate feature importances for average income in 2020. This approach provided a ranking of the most influential factors such as degree rate, immigrant rate, lone parent rate, and population.

Analysis

We begin the analysis by first examining the 2020 and 2015 average income after tax.



West Vancouver has the highest income, and Chilliwack has the lowest income in 2015. In 2020, Richmond falls one place down and becomes the lowest income.

AT: After tax

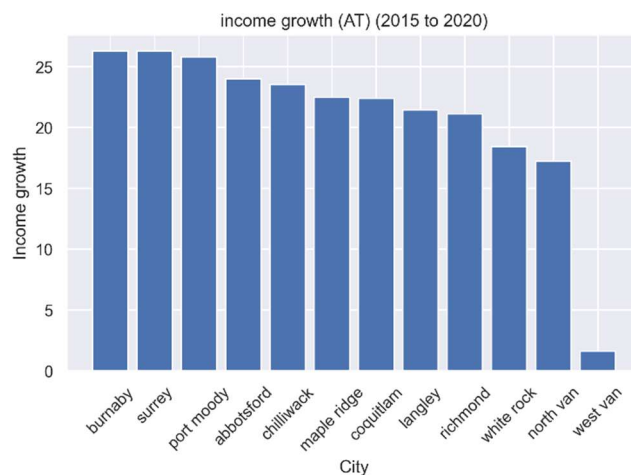
Rate: $x / \text{population}$

Note: only average income AT was used, so all mentions of 'income' refers to that.

Degree rate = Bachelor rate = Bachelor or above = number of people with bachelor or higher.

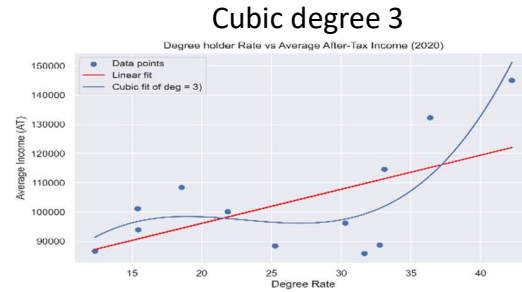
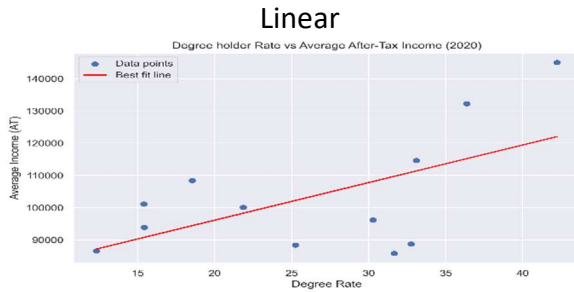
The highest income (West Vancouver) earned 103.57% more than the lowest income, Chilliwack in 2015. This difference was then significantly reduced in 2020. The highest income (West Vancouver) and the lowest income (Richmond, no longer Chilliwack) have a difference of 68.99% in 2020. That is supposedly a growth of 34.58% for the lowest income city.

But that is not the full picture. Let's look at income growth rates for each city from 2015 to 2020.



West Vancouver's income growth is only 1.59%, compared to an average of 20.8% across all 12 cities, a strikingly low growth rate. This could suggest two possibilities: either earnings are being underreported, or the city is already near the upper limit of achievable incomes. Notably, about one-third of West Vancouver's residents are business owners, compared to a regional average of one-fifth. This higher share of self-employed individuals may provide greater opportunity to underreport earnings. For this reason, I also compared lowest income city to the second-highest earning city, North Vancouver. In 2015, North Vancouver's average income was 60.86% higher than the lowest income city, but by 2020, this gap had narrowed to 54.07% which still implies that lower income cities are somewhat catching up. What factors might explain this shift?

To determine whether there was a statistically significant income growth from 2015 to 2020, I conducted a Mann-Whitney U test, as the small sample size and non-normal distribution made a t-test inappropriate. The p-value for the Mann-Whitney test is 0.01202 which is smaller than our threshold of 0.05. This means that there is in fact a difference between 2015 to 2020 taxed average incomes. Now, we can explore what has potentially caused this shift. I began by using the seemingly obvious factor, the number of degree holder's (rate) vs average income, then I fit a regression line (OLS) to it.

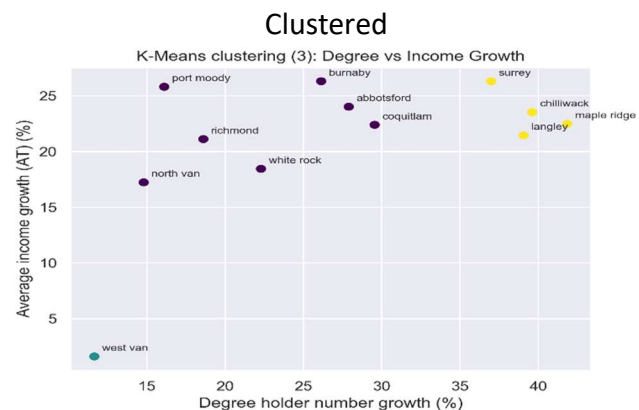
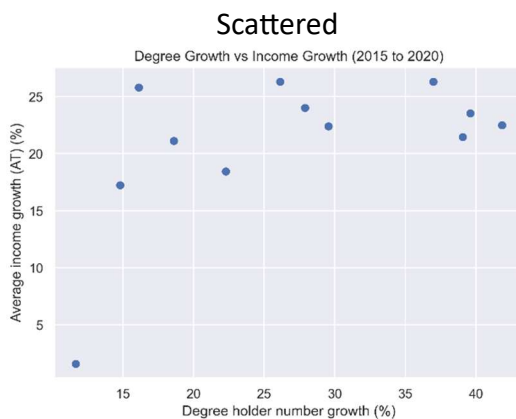


The linear regression yielded a p-value of 0.043, below the 0.05 significance threshold, indicating a statistically significant positive relationship between the proportion of degree holders and the average after-tax income. The correlation coefficient was approximately 0.6, suggesting a moderately strong association. However, because the relationship is not strictly linear, I also plotted a third-degree polynomial curve, which appeared to fit the data more closely.

The final dataset also includes several other variables worth exploring. For example, the correlation between the proportion of immigrants and average income is only 0.0125, indicating virtually no relationship. The correlation between income growth and population growth is 0.258, suggesting a modest positive relationship. Examining lone-parent households, the correlation with average income is just 0.040, meaning lower-income cities do not necessarily have more lone-parent households. However, there is a very strong correlation (0.987) between population size and the number of lone-parent households which is an unsurprising result, as larger populations naturally have higher counts across most demographic categories. The correlation between degree rate growth and income growth is noteworthy at 0.533 which suggests a moderate to strong positive relationship. This value goes up to almost 0.61 without West Vancouver.

Clustering:

To prepare for clustering, I first aligned the 2015 and 2020 datasets (previously separated). I then plotted degree rate growth on the x-axis and average income on the y-axis, allowing us to identify cities that experienced high growth in both metrics.



The K-means clustering of degree-holder growth versus average income growth reveals three distinct groupings of cities. One cluster, represented by West Vancouver, shows both low degree growth and low-income growth, standing apart from all others. A second cluster, including cities such as Port Moody, Burnaby, Abbotsford, and Coquitlam, demonstrates moderate degree growth paired with relatively high-income growth. The third cluster, containing Surrey, Chilliwack, Langley, and Maple Ridge, exhibits both high degree growth and high-income growth.

This plot suggests that growth in educational accomplishments does play a role in income growth. However, the relationship is not entirely consistent across cities. While some municipalities with strong degree growth also see high income gains, others achieve similar income growth with more modest increases in degree rates, suggesting that additional factors could be affecting our income growth as well.

Feature importance

To identify which factors most strongly influence income, I used feature importance scores from a Random Forest regressor. Since the goal was not to build a predictive model, the data was not split into training and testing sets. Instead, the model was applied to the full dataset to assess variable importance. Among the features considered, the proportion of degree holders and the lone-parent rate were shown to be the most influential. While earlier results showed no strong linear correlation between lone-parent rate and income, this finding suggests there may be a nonlinear or interaction-based relationship between the two.

Final words

This analysis of Canadian census data for 12 cities in British Columbia shows that income growth between 2015 and 2020 was statistically significant but uneven. Lower-income cities appear to be narrowing the gap with higher-income cities, as seen in the reduced difference between the highest- and lowest-earning municipalities. West Vancouver stands out for its extremely low growth rate despite being the wealthiest city, which may show underreporting or limits to further income increases at the top.

Education plays an important role in income growth, with increases in degree rates generally linked to higher average incomes. However, the relationship is not fully consistent, and some cities gain income without matching growth in educational attainment. Other variables, such as immigration rates and lone-parent household rates, show weaker or more complex patterns. Random Forest analysis highlighted lone-parent rate as the second-most important factor after degree rate, suggesting nonlinear or interaction effects that simple correlations cannot capture. Overall, education appears central to income growth, but other social and demographic factors also influence outcomes. In conclusion, evidence suggests that several lower-income cities are narrowing the gap, and increased academic education appears to be a key driver of this growth.

Other Cities Slowly Narrowing the Gap to the Top Two (NV and WV)

3 of the lowest income cities are among the highest income growth marked with (*).

<i>City</i>	<i>Growth (2015 to 2020)</i>
Burnaby *	26.3%
Surrey	26.2%
port moody	25.8%
Abbotsford *	24.0%
Chilliwack *	23.5%
maple ridge	22.4%

Average growth among
all 12 cities: 20.8%

Limitations

Going into this project, I expected to find more concrete relationships between the variables I chose. While I managed to do that with some, others were affected by noise and uncertainty. I had hoped to include spoken language as another variable, but it was impossible to draw any meaningful conclusions from it. The dataset was limited to 12 (total of 24 over 2015-2020) cities, and just gathering, cleaning, and making sure I was getting the right values took much longer than I thought.

One big thing I wanted to show was whether higher-income or lower-income households had a strong relationship with income growth, but the data I had didn't allow me to prove it directly (although the feature importance did suggest some kind of connection).

The datasets from different years didn't follow the same format, so I had to redo some of the cleaning and sometimes manually double-check that I was getting the right numbers. There were also many metrics with very similar topic names but slightly different meanings, which made things tricky. And in some cases, the limited amount of data kept me from producing concrete results. External influences were also not in dataset such housing market shifts or industry shifts.

Data source:

- <https://www12.statcan.gc.ca/census-recensement/2021/dp-pd/prof/search-recherche/lst/results-resultats.cfm?Lang=E&GEOCODE=59>
- <https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/search-recherche/lst/results-resultats.cfm?Lang=E&TABID=1&G=1&Geo1=&Code1=&Geo2=&Code2=&GEOCODE=59&type=0>